When Do Benchmarks Predict LLM Adoption?

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Abstract

While leaderboard scores drive substantial investment, their ability to predict the uptake of models within the scientific community remains empirically untested. We introduce a dataset pairing 224 LLMs with scientific adoption signals extracted from 48,190 papers via sentence-level classification and calibrated with adjusted classify-and-count and inverse-probability weighting. 63 of these are matched to scores on the Hugging Face Open LLM Leaderboard (OLLM; v1, v2, or both). For 2024 open-weight releases, all five benchmarks exhibit positive Spearman rank correlations with one-year scientific adoption ($\rho \in [0.53, 0.74]$, FDR-adjusted $q \leq 0.01$) that survive controlling for parameter count, and four of five also survive jointly controlling for organizational reputation (frontier-lab indicator). For the two benchmarks with a 2023 cohort (BBH and MATH Lvl 5), correlations are statistically indistinguishable from zero, with confidence intervals consistent with effects up to $|\rho| \approx 0.5$ in either direction. Two leaderboard-internal diagnostics (within-cohort score variance and score–size collinearity) help characterize three failure modes: size-redundancy, floor compression, and saturation. The cross-cohort divergence is consistent with the benchmark–adoption correlation varying by (benchmark, cohort) pair. The datasets and code are released to support studies of LLMs, benchmarks, and downstream adoption at: `https://anonymous.4open.science/r/benchmark-validity-AC28`

1 Introduction

Benchmark leaderboards are the primary mechanism by which the field ranks large language models (LLMs). Benchmarks are especially valuable when they provide signal about downstream outcomes such as real-world utility or model selection behavior, but this link is rarely tested directly. If leaderboard scores only weakly predict downstream selection, the field may be optimizing signals with limited or context-dependent practical relevance. If they do, benchmark design becomes consequential, as models that exploit evaluation weaknesses may be preferentially deployed because practitioners rely on these scores. This raises a central question: to what extent do leaderboard rankings predict downstream adoption?

Answering this question is challenging on both sides of the relationship. Benchmark scores are affected by well-known issues such as Goodhart's law, contamination, and score compression as models saturate tasks [21]. Model size may confound the relationship between benchmark scores and adoption, as larger models are more capable and widely used, which makes the model metadata (often incomplete) critical. On the adoption side, deployment data is often proprietary, and commonly used public proxies (e.g., download counts) conflate sustained use with transient interest. Scientific adoption provides a more deliberate signal: when a model is used in a published study, it reflects a consequential choice. However, extracting this signal at scale requires distinguishing genuine usage from incidental mentions in text. We construct a dataset of 224 LLMs paired with adoption signals

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from text analysis of 48,190 scientific papers, classify each citation as genuine adoption or contextual mention, and link these signals to scores on OLLM for the 63 models with available scores (Figure 1).

We find that, for models released in 2024, benchmark scores consistently rank-correlate with one-year post-release scientific adoption across all five benchmarks considered ($\rho \in [0.53, 0.74]$, FDR-adjusted $q \leq 0.01$), and these relationships persist after controlling for model size. In contrast, for models released in 2023, two benchmarks show no measurable association with adoption. These differences are consistent with two identifiable patterns in the leaderboard data. First, redundancy: in some cases, benchmark scores are highly collinear with model size and provide little additional information once size is accounted for. Second, score compression: when within-cohort score variance is low, benchmarks lack the resolution needed to distinguish between models. Both patterns are directly observable from leaderboard statistics. Together, these results indicate that the relationship between benchmark performance and adoption is not uniform across settings. Benchmark scores can provide meaningful signals of downstream use, but their interpretability depends on properties of the evaluated model population and the structure of the benchmark itself.

The 2023 and 2024 models are particularly significant as early cohorts in the modern open-weight LLM era. Following the releases of LLaMA [50] in February 2023 and Mistral 7B [32] in September 2023, downloadable and modifiable models became a viable substrate for research, and adoption became observable as a selection process potentially influenced by benchmark leaderboards. As a result, these cohorts provide an empirical baseline for analyzing subsequent developments. Many current debates in the field, including those on benchmark validity, leaderboard design, and model evaluation more broadly, rest on assumptions that have not yet been tested on these formative cohorts. The present moment, with the 2024 cohort sufficiently mature to exhibit adoption signals and OLLM still an active measurement instrument, is the first point at which such a test becomes feasible. Against this backdrop, our contributions are as follows:

- **A dataset of 224 LLMs (released 2022–2024)** with classifier-corrected scientific adoption estimates derived from 48,190 scientific papers, linked where possible to OLLM v1/v2 scores. The dataset includes manually-curated LLM metadata, including size, organization, external crosswalks, and it is structured to enable studies of LLMs, benchmarks, and adoption beyond the one we report.
- **Benchmark scores are positively associated with scientific adoption.** For 2024 open-weight model releases, all five OLLM v2 benchmarks are positively rank-associated with 12-month scientific adoption beyond what parameter count alone explains ($\rho \in [0.53, 0.74]$, $q \leq 0.01$); four of five (MATH Lvl 5, MuSR, GPQA, BBH) also survive joint control for size and a frontier-lab indicator at $p < 0.05$, with IFEval marginal at $p < 0.10$.
- **Three empirically observable failure modes for the benchmark–adoption correlation**. We observe size-redundancy, floor compression, and saturation as modes under which leaderboard scores cease to predict adoption. Each is diagnosable from leaderboard data alone via two statistics: within-cohort score variance and score–size collinearity.

We study the relationship between benchmark scores and scientific adoption for LLMs released in the 2023-2024 cohorts. Rather than interpreting scientific adoption as a direct measure of model quality, we treat it as a downstream selection outcome shaped jointly by model performance and broader ecosystem factors. The released dataset spans models from 2022-2024 and supports analogous evaluations across multiple leaderboards over this period. Extensions to later cohorts and to non-academic forms of adoption are left for future work. More broadly, we frame this as a benchmark evaluation problem: we ask how much information widely used LLM benchmark scores carry about downstream scientific adoption, and we release a reusable dataset to support future analyses.

1.1 Related work

Benchmark validity and evaluation methodology. A substantial body of work questions whether standard benchmarks measure the constructs they claim, from general NLP [10, 12, 48] to preference alignment [59] and clinical reasoning [3], with broader concerns about real-world utility [7]. Static benchmarks also saturate over time [2], eroding discrimination between models and incentivizing dataset-specific optimization [14, 55]; multidimensional frameworks have been proposed in response [36]. At the item level, Truong et al. [52] identify broken items across nine benchmarks

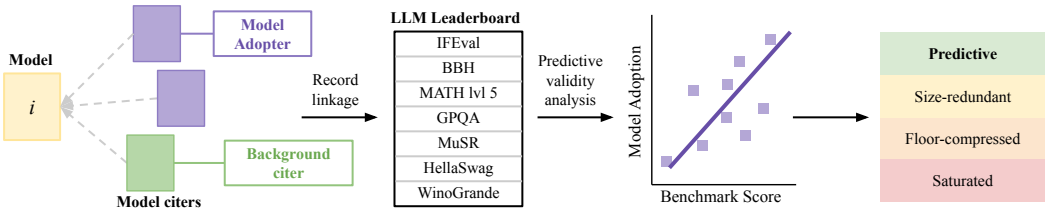


Figure 1: Illustration of the Benchmark Score–Adoption analysis. Models' data is supplemented with metadata, citation-based adoption measurements, and benchmark scores. We analyze the association between benchmark scores and adoption, and the mechanisms underlying this association.

(up to 84% precision), and Mousavi et al. [39] report 22–30% flawed items in three reasoning benchmarks. This literature establishes that benchmarks are noisy and prone to saturation, but focuses on *internal* validity. We instead ask how much *external* predictive signal survives, and characterize three observable failure modes, i.e., redundancy, compression, and saturation, that erode it.

Scaling laws and latent capability structure. A second line of work shows that performance differences across models are largely explained by scale-related variables, such as parameter count, training compute, dataset size [31, 33]. Empirical scaling laws demonstrate smooth, predictable improvements as a function of these resources [31, 33]. Diverse evaluation tasks are correlated [13], suggesting leaderboard performance reflects a small number of latent factors rather than task-specific competencies; even safety benchmark variance is largely driven by capability and compute [43]. This raises the possibility that benchmark scores are partially redundant with scale, motivating our analysis of whether benchmark–adoption relationships persist after controlling for model size.

Adoption and downstream impact. A third body of work studies how scientific and technological artifacts diffuse and generate impact. In machine learning, adoption is proxied via application usage [57], citation patterns [27], or system integration [17], while science-of-science research models long-term influence [24]. These studies show that uptake reflects both intrinsic quality and visibility, but whether benchmark scores predict real-world use remains untested. Empirical work on foundation model adoption [27, 17, 51] finds that uptake is shaped by compute access, openness, and ecosystem maturity. A contemporaneous preprint [51] applies a similar classification-based pipeline across modalities without releasing data, code, or crosswalks. We provide the first formal data release for this line of work, with an OLLM v1/v2 crosswalk, classifier-corrected adoption counts with calibrated uncertainty, and predictive tests against leaderboard scores. A parallel literature audits evaluation artifacts at scale, documenting reporting and provenance issues [9, 56] and formalizing benchmark lifecycle criteria [44]. These studies focus on benchmark design and documentation, whereas we study *predictive* validity, releasing a dataset of 224 models with adoption signals (63 linked to OLLM scores) and providing direct evidence that 2024 leaderboard scores correlate with one-year post-release uptake.

2 LLM Benchmark-Adoption Dataset

2.1 Dataset overview

We introduce a unified dataset of LLM adoption, benchmark performance, and bibliographic metadata, linking a curated set of 224 LLMs from the Epoch AI Index with scientific usage signals derived from the Semantic Scholar Academic Graph (S2AG) [35, 54]. The dataset integrates three core components. First, a curated model registry (`models`) containing metadata on model size, organization, release date, and openness. Second, a citation-derived adoption layer (`adoption`) constructed from sentence-level classification of 48,190 retrieved citing papers (out of 201,229 total citations to these models in S2AG); each sentence is classified to distinguish actual model usage from contextual mention. Third, benchmark performance data from OLLM v1 and v2 leaderboards (`leaderboards`), covering 11 evaluation benchmarks including IFEval [60], MuSR [47], GPQA [42], Big Bench Hard (BBH)[49],

MATH Lvl 5 [30], ARC [15] and HellaSwag [58], MMLU [29], WinoGrande [45], GSM8K [16], and TruthfulQA [37].

Interoperability and Extensibility. We provide crosswalks linking model identifiers to Semantic Scholar and Hugging Face/OLLM identifiers (`crosswalks`), enabling integration with external bibliographic and leaderboard metadata. This enables on-demand enrichment with publication venue, authorship, affiliations and external identifiers (e.g., via DOI) via the S2 API (for both models and citing papers). Beyond this analysis, the dataset is designed to support studies of LLM diffusion, geographic and institutional adoption patterns, domain-specific uptake, among other research questions.

Tools and Resources. The release is packaged with a Croissant metadata file [1] and a lightweight Python module (`benchmark_validity`) supporting temporal aggregation of daily-resolution adoption signals (e.g., to monthly, quarterly, yearly resolution). This module also propagates uncertainty from the citation classification pipeline to aggregated statistics. In addition, we include a reproducibility package (`notebooks`) that fully regenerate the main results and serve as a demonstration of data use and aggregation (see Appendix **??** for details).

2.2 Dataset Construction

Model selection. We exported all available models from the Epoch AI Index [20, 46] (March 2025 snapshot) as our initial candidate pool and apply the following selection criteria. First, the model must be classified as an LLM in the Epoch database. Second, to be used as a scientific artifact, the model must have been *publicly accessible* at the time of adoption, whether through open weights or a public API (API-only models without publicly available weights are subsequently excluded at the OLLM matching step below). Therefore, internally developed models without a public release (e.g., Chinchilla, Flamingo) are excluded. Third, the model must be a *deployable artifact* rather than a methodological contribution: we exclude attention mechanism variants (e.g., ScatterBrain, Linear Transformer), compression techniques (e.g., SqueezeBERT, MobileBERT), and training infrastructure systems (e.g., Megatron-LM), as scientists cite these as techniques rather than adopting them as research tools. We manually supplement each entry with metadata, such as model size (total parameters) and availability (e.g., open weights, API).

Citation-based adoption classification. Model adoption is inferred by extracting and classifying text from its citing papers. We map the selected models to the Semantic Scholar Academic Graph (S2AG) [35, 54] (February 2026 snapshot) to track citation counts. We retrieve full text of citing papers from S2ORC corpus [38] where available and supplement it through a custom-made pipeline that retrieves papers from other open sources (arXiv, PubMed Central). The 224 large-language models in our sample are cited 201,229 times across the literature; we retrieve full text for 48,190 of these citing papers. For each citing paper, we extract citation sentences referencing the model and classify as *adoption* (the citing paper applies, fine-tunes, or extends the model) or *context* (background or comparative reference) via zero-shot prompting of GPT-4.1-mini [40], with model-variant disambiguation via Llama-3.1-8B [19]. A paper is labeled as adopting the model if any of its citation sentences is classified as adoption. The classifier achieves macro-F1 of 0.78 on a held-out validation set; per-class metrics and prompts are in Appendix A. We estimate adoption counts by applying Adjusted Classify-and-Count [23] (a methodology with recent application to LLM-use prevalence estimation [53]) to correct for classifier bias (Appendix B), then scaling to the full citing population via inverse-probability weighting to account for non-random retrievability (Appendix C). We propagate sampling variability via paper-level resampling within each model (preserving the joint distribution of predicted labels and IPW weights) and classifier calibration uncertainty via Beta posteriors on $(\widehat{\mathrm{TPR}}, \widehat{\mathrm{FPR}})$, combining both in a bootstrap with $B = 10{,}000$ draws (Appendix D).

Benchmark Data and Matching. We retrieve scores from both OLLM v1 [6] and OLLM v2 [26] (April 2026 snapshot). Because OLLM requires publicly available model weights for evaluation, this matching step implicitly restricts coverage to open-weight models, whereas our dataset includes both open-weight and API-only models. We merge these with our model adoption data using a combination of fuzzy string matching and manual review. We select a canonical official release variant using predetermined matching rules based on organization, version, and parameter count;

when multiple entries remain, we choose the highest-scoring non-flagged entry. Benchmark scores of exactly zero were treated as missing data. This rule was applied uniformly across benchmarks because exact zero scores for the matched modern LLMs are operationally implausible and likely reflect failed evaluation runs rather than true performance.

3 Benchmark Score–Adoption Association

We stratify models by release year to isolate within-cohort signal from cross-cohort variation in model capabilities, training regimes, and adoption dynamics. Each cohort consists of models released in a single calendar year and is analyzed separately. We include only benchmark-cohort pairs where the benchmark was available prior to the model's adoption window. We retain cohorts with at least ten models with non-missing benchmark scores [11]. Our primary analysis uses OLLM v2, as OLLM v1 was retired by Hugging Face in mid-2024 [25].

We analyze selected OLLM v1 benchmarks to characterize saturation-related patterns associated with benchmark retirement. This scope follows the published OLLM v1 deprecation rationale [25], which identifies retirement due to saturation (HellaSwag, WinoGrande, ARC), contamination from training-data overlap (GSM8K, TruthfulQA), and benchmark errors (MMLU). Our analysis focuses on score-distribution patterns associated with saturation; contamination and implementation or measurement errors represent distinct validity concerns outside the scope of this study.

We use the model adoption count during each model's first 12 months after release as the primary outcome. A fixed per-model observation window is used to make adoption comparable across release years, and 12 months captures the initial wave of scientific uptake. All models in our sample are fully observed within this window (released by end of 2024, citations tracked through end of 2025).

3.1 Primary Analysis

For each benchmark b with benchmark score $S_{b,i}$ on model i and adoption count y_i, we report the Spearman rank correlation $\rho^b_{s,a} = \mathrm{Spearman}(S_{b,i}, y_i)$. We adopt $\rho^b_{s,a}$ as our primary measure of association as it captures any monotonic relationship between benchmark performance and adoption. This is important given heterogeneity in benchmark scale, ceiling effects, and distributional shape, with no a priori reason to expect linearity. We compute 95% confidence intervals via bootstrap resampling ($N_{\mathrm{boot}} = 10{,}000$) and two-sided p-values via within-cohort permutation ($N_{\mathrm{perm}} = 10{,}000$). See Appendix D for details.

Robustness checks. We report Benjamini–Hochberg adjusted q-values alongside p-values, controlling the false discovery rate at 5% across benchmark-level tests [8]. We assess sensitivity to influential observations via a trimming analysis that removes the highest- and lowest-scoring model and recomputes correlations for all benchmarks with at least one significant association ($q < 0.10$).

3.2 Mechanism Analysis

A raw rank correlation between benchmark score and adoption may reflect information specific to the benchmark, or it may simply track a simpler observable such as model size. To disentangle these effects, we report additional rank statistics and within-cohort score variance.

Score–size collinearity. We compute $\rho^b_{s,z} = \mathrm{Spearman}(S_{b,i}, z_i)$, where $z_i = \log(\mathrm{params}_i)$, to quantify the extent to which benchmark ranking aligns with model size.

Size-adoption association. We report $\rho_{z,a} = \mathrm{Spearman}(z_i, y_i)$, the rank correlation between size and adoption. Together with $\rho^b_{s,a}$ and $\rho^b_{s,z}$, this helps disentangle whether adoption is more closely aligned with model size, benchmark performance, or signals shared between the two.

Partial Spearman correlation. To isolate the predictive content of benchmark rank that is orthogonal to model size, we compute the partial rank correlation $\rho^b_{s,a|z}$, defined as the Pearson correlation

between the residuals of $R(S_{b,i})$ and $R(y_i)$ after each is regressed on $R(z_i)$. This is algebraically equivalent to $\rho^b_{s,a|z} = \frac{\rho^b_{s,a} - \rho^b_{s,z}\,\rho_{z,a}}{\sqrt{\left(1-(\rho^b_{s,z})^2\right)\left(1-\rho^2_{z,a}\right)}}$.

The partial correlation is the complementary statistic for our central claim: a benchmark adds predictive value beyond model size only when $\rho^b_{s,a|z}$ is materially nonzero. When $\rho^b_{s,a}$ is high but $\rho^b_{s,a|z}$ is near zero, the benchmark primarily reflects model size rather than providing independent signal. We compute two-sided p-values for $\rho^b_{s,a|z}$ via residual permutation [34, 4], which preserves the dependence structure with z_i under the null of no partial association. See Appendix D for details.

Robustness to organizational reputation. As a robustness check, we extend the partial correlation to jointly control for log-parameters and a binary frontier-lab indicator, computing $\rho_{s,a|z,f}$ using the same residual permutation procedure [1]. The frontier-lab indicator is used as a coarse control for organizational reputation and ecosystem advantage, not as a full measure of institutional prestige or reach.

Within-cohort score variance. We additionally report the within-cohort variance of benchmark scores σ^2_b as a diagnostic for compression. A benchmark with low σ^2_b has limited discriminative power; in this regime, low $\rho^b_{s,a}$ reflects score compression rather than invalidity, distinguishing it from benchmarks that are merely redundant with size (low partial correlation).

4 Results

We consider a benchmark predictive of adoption within a cohort when it exhibits a positive and statistically significant association ($\rho^b_{s,a} > 0$, $q < 0.05$). Robustness is assessed by examining whether this association persists under trimming and when controlling for model size. Table 4.1 summarizes the results, with sensitivity analyses in Appendix E.

4.1 Benchmark scores predict scientific adoption

Across all five OLLM v2 benchmarks [26], the 2024 cohort exhibits a positive and statistically significant rank correlation between benchmark performance and downstream adoption ($\rho_{s,a} \in [0.53, 0.74]$, all $p < 0.01$, $q \leq 0.01$), with confidence intervals excluding zero. The strongest associations are concentrated in GPQA and MuSR, both recent reasoning-focused benchmarks. We re-estimate these relationships using partial Spearman correlations controlling for log-parameters. The partial correlations are comparable in magnitude to the marginal effects across most benchmarks. We further control jointly for log-parameters and a binary frontier-lab indicator. Under this specification, four of five benchmarks retain a positive and statistically significant association with adoption: MuSR ($\rho_{s,a|z,f} = 0.66$, $p < 0.001$), GPQA (0.59, $p < 0.001$), MATH Lvl 5 (0.69, $p < 0.001$), and BBH (0.43, $p < 0.05$). IFEval attenuates to $\rho_{s,a|z,f} = 0.40$ and is marginally significant ($p < 0.10$).

4.2 Failure mode I: size-collinearity

Within the 2023 cohort, BBH score is tightly coupled to model size: Spearman $\rho_{s,z} = 0.75$ between score and log-parameters, with OLS yielding $R^2 = 0.68$ and $\beta = 17.1$ ($p < 0.001$, Figure 2a). Size itself is weakly associated with adoption in this cohort ($\rho_{z,a} = 0.10$), so the benchmark's collinearity with size links it to a variable that is itself weakly predictive of adoption. The marginal score–adoption correlation is accordingly near zero ($\rho_{s,a} = -0.06$). After controlling for log-parameters, the partial correlation is $\rho_{s,a|z} = -0.21$, indicating no remaining positive association between score and adoption within size strata.

The 2024 cohort represents a distributional break rather than a change in benchmark design. Efficient sub-10B parameter models (including several released in 2024) now span the full BBH score range previously occupied only by $\geq$30B models, and several sub-10B 2024-cohort models exceed all 2023-cohort models regardless of size (Figure 2b). The OLS coefficient drops to $\beta = 11.0$ and explanatory power falls from $R^2 = 0.68$ to $R^2 = 0.34$, indicating reduced dependence of score

[1] Meta, Google/DeepMind, Mistral, OpenAI, Anthropic, Microsoft, Alibaba/Qwen

Benchmark	Cohort	N	Marginal		Partial (z)	Partial (z, f)				
			$\rho_{s,a}$	q_ρ	$\rho_{s,a\|z}$	$\rho_{s,a\|z,f}$	$\rho_{z,a}$	$\rho_{s,z}$	σ^2	$\bar{S}$
MuSR (2023)	2024	24	0.74*** [0.50, 0.87]	0.00	0.74*** [0.47, 0.88]	0.66*** [0.27, 0.85]	0.22	0.46	0.002	0.113
GPQA (2023)	2024	24	0.68*** [0.38, 0.84]	0.00	0.72*** [0.39, 0.87]	0.59** [0.17, 0.81]	0.22	0.64	0.004	0.093
MATH Lvl 5 (2021)	2024	23	0.72*** [0.41, 0.88]	0.00	0.77*** [0.46, 0.90]	0.69*** [0.30, 0.86]	0.22	0.65	0.014	0.151
	2023	18	-0.08 [-0.55, 0.38]	0.82	-0.14 [-0.58, 0.32]	-0.06 [-0.53, 0.35]	0.10	0.44	0.001	0.035
BBH (2022)	2024	24	0.57** [0.24, 0.76]	0.01	0.61*** [0.24, 0.79]	0.43* [0.01, 0.69]	0.22	0.74	0.027	0.327
	2023	18	-0.06 [-0.53, 0.45]	0.82	-0.21 [-0.66, 0.27]	-0.22 [-0.66, 0.25]	0.10	0.75	0.014	0.214
IFEval (2023)	2024	24	0.53** [0.16, 0.79]	0.02	0.50* [0.11, 0.79]	0.40† [-0.07, 0.75]	0.22	0.54	0.043	0.556
WinoGrande (2019)	2024	11	-0.14 [-0.66, 0.51]	0.82	-0.14 [-0.71, 0.60]	-0.35 [-0.86, 0.59]	-0.17	0.00	0.007	0.751
HellaSwag (2019)	2024	11	-0.22 [-0.73, 0.46]	0.76	-0.24 [-0.77, 0.53]	-0.42 [-0.88, 0.57]	-0.17	-0.07	0.008	0.788

Table 1: Benchmark–adoption correlations. Outcome: log cumulative adoptions one year post-release. Columns report marginal Spearman correlation ($\rho_{s,a}$), partial correlations controlling for log-parameters ($\rho_{s,a|z}$) and additionally for a frontier-lab indicator ($\rho_{s,a|z,f}$), size-adoption correlation ($\rho_{z,a}$), and score-size correlation ($\rho_{s,z}$). Within-cohort score variance (σ^2) and mean score ($\bar{S}$) summarize score distribution. 95% bootstrap confidence intervals in brackets; significance codes: $\dagger p < 0.10$, $* p < 0.05$, $** p < 0.01$, $*** p < 0.001$.

on size. With score carrying information orthogonal to size, the partial correlation increases to $\rho_{s,a|z} = 0.61$ ($p < 0.01$) and the marginal to $\rho_{s,a} = 0.56$. The parameter-count distribution of adopted models also broadens between 2023 and 2024 (Figure 3).

4.3 Failure mode II: floor compression

MATH Level 5 exhibits a different failure mode in 2023 than BBH: rather than collinearity with model size, it shows range restriction. Within-cohort score variance is $\mathrm{Var}(s) = 0.001$ (on a $[0, 1]$ scale), more than an order of magnitude below the 2024 value of 0.015 (Table 4.1). The 2023 distribution is compressed against the floor: most models score in the 0–5 range, with a single outlier near 10 (Figure 2c), and even this narrow spread is largely unexplained by model size ($R^2 = 0.14$). In this regime, both marginal and partial correlations are near zero ($\rho_{s,a} = -0.14$; $\rho_{s,a|z} = -0.14$), reflecting limited usable variation rather than a strong size relationship. The benchmark therefore provides little discriminative signal in 2023 because the model population lies outside its effective measurement range.

The 2024 cohort differs from this regime, coinciding with advances in efficient training (particularly parameter-efficient fine-tuning and improved data curation) that allow sub-10B models to achieve non-trivial performance on hard mathematical reasoning [18, 28]. The score distribution expands, with within-cohort variance increasing to 0.015 and scores spanning the 0–40 range. Under this expanded support, MATH Level 5 recovers a strong benchmark–adoption correlation ($\rho_{s,a} = 0.72$, $p < 0.001$), with partial correlation remaining similar after controlling for size ($\rho_{s,a|z} = 0.77$, $p < 0.001$), indicating that benchmark scores carry information beyond scale once sufficient variation is present.

4.4 Failure mode III: saturation

A third failure mode is consistent with saturation, which mirrors floor compression at the upper end of the score range. As model populations approach a benchmark's ceiling, within-cohort score variance collapses and the benchmark loses discriminative power. For the five OLLM v2 benchmarks

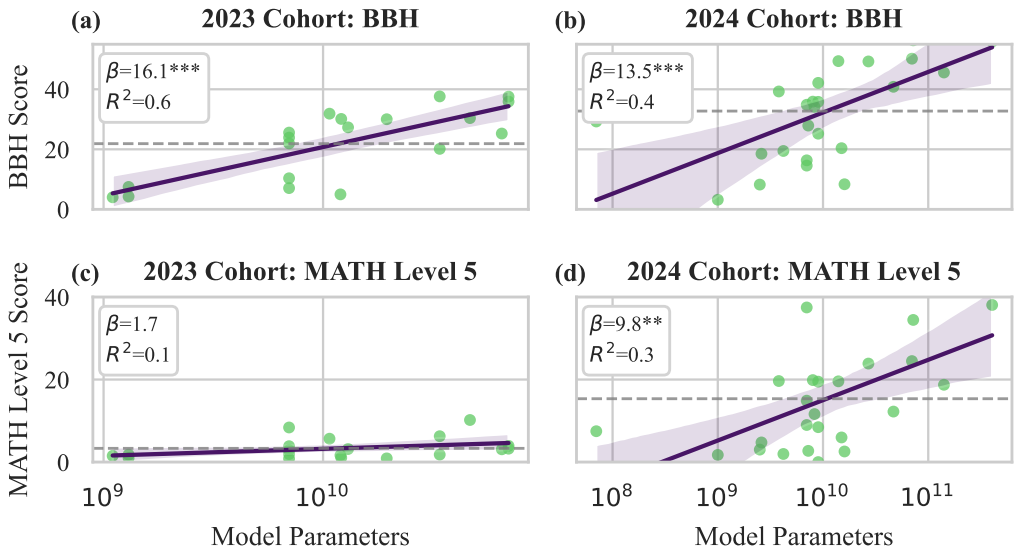


Figure 2: **Two failure modes diagnosed via score–size relationships.** Each panel plots benchmark score vs. $\log_{10}$(params). **(a)** BBH 2023: score tightly determined by size ($\rho_{s,z} = 0.75$) - size-collinearity failure. **(b)** BBH 2024: score–size coupling substantially reduced. **(c)** MATH Lvl 5 2023: scores compressed near zero ($\mathrm{Var}(s) = 0.001$ on $[0, 1]$) — floor-compression failure. **(d)** MATH Lvl 5 2024: variance recovers ($\mathrm{Var}(s) = 0.015$). Solid lines: OLS overlay (descriptive only). Dashed lines: cohort mean score.

in the 2024 cohort, mean scores range from $\bar{S} = 0.09$ (GPQA) to 0.55 (IFEval), all well below the score ceiling (Table 4.1). Within-cohort score variances ($\sigma^2 \in [0.002, 0.043]$) reflect heterogeneity in model performance rather than ceiling saturation.

Two OLLM v1 benchmarks in the same 2024 cohort illustrate saturation. HellaSwag compresses to $\bar{S} = 0.79$ (std 0.09) and WinoGrande to $\bar{S} = 0.75$ (std 0.08), with markedly reduced dispersion relative to the 2023 cohort (HellaSwag std 0.17; WinoGrande 0.16). Score–adoption correlations are null ($\rho_{s,a} = -0.22$ and -0.14; both $p > 0.5$) and remain null after controlling for model size ($\rho_{s,a|z} = -0.24$ and -0.14).

5 Discussion

Our findings provide empirical evidence that traditional LLM leaderboards contain meaningful predictive signal for real-world scientific adoption under specific cohort and evaluation conditions. This relationship is contingent on benchmark-population alignment and may vary across time and model regimes. Our results have implications for multiple stakeholders. For LLM developers, leaderboard rankings may serve as predictive signals for model release, positioning, and investment decisions. For the AI safety and alignment community, the results underscore the importance of preventing leaderboard gaming, given the downstream consequences that benchmark performance may have for model adoption. For the evaluation and datasets community, our findings support the view that benchmarks can function as valid and valuable measurement instruments, while also motivating the development of improved benchmarks and use of lightweight diagnostic checks that can be applied before substantial adoption signals emerge. Overall, these results suggest that leaderboard signal is not a static property

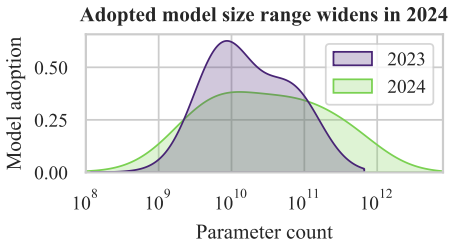


Figure 3: Adopted model parameter-count distribution widens between 2023 and 2024 cohorts. Density: Gaussian KDE.

of benchmarks, but an emergent property of the interaction between evaluation design, the distribution of evaluated models, and the evolving ecosystem of model development and adoption.

Implications for leaderboard design. The failure modes identified above are directly detectable from leaderboard-internal statistics, without requiring external adoption data. Score-size collinearity flags cases where benchmark rankings primarily reflect model scale rather than independent signal, while low within-cohort score variance indicates compression or saturation that eliminates discriminative power. Together, these diagnostics provide lightweight checks of whether a benchmark contributes information beyond model-card metadata. Importantly, this redundancy is distinct from classical confounding: even when controlling for parameter count, a benchmark may retain predictive association with adoption while offering no additional informational gain beyond model size. In this regime, the benchmark is not invalid as a capability measure, but effectively redundant with existing model metadata. This condition is reversible—when score-size coupling weakens, conditional predictive signal re-emerges (e.g., $\rho_{s,a|z} = 0.61$ for BBH in 2024). For most benchmarks, predictive signal persists after controlling for both model scale and organizational reputation, albeit with heterogeneous attenuation across tasks. These diagnostics thus operationalize a distinction between benchmarks that add independent rank information and those that restate model scale in disguise, enabling prospective monitoring of leaderboard validity as model populations evolve.

Frontier-lab mediation. The differential attenuation pattern is itself indicative. MATH Lvl 5 and MuSR exhibit the smallest and nearly identical attenuation ($0.77 \rightarrow 0.69$ and $0.74 \rightarrow 0.66$; 10–11% relative reductions), indicating that their score-adoption association is comparatively robust to controls for organizational reputation (performance on hard mathematical reasoning and multi-step soft reasoning predicts adoption regardless of who released the model). GPQA shows moderate attenuation (18%). BBH attenuates most substantially ($0.61 \rightarrow 0.43$, roughly 30%), suggesting that its 2024 score–adoption signal is more sensitive to controls for frontier-lab provenance. IFEval shows similar attenuation (20%) and falls below conventional significance under double control. The pattern is consistent with heterogeneity in how different benchmarks align with models released by frontier versus non-frontier organizations, rather than a uniform relationship across task types.

Interpretation against prior debates. Our findings complement the literature on benchmark construct validity and predictive validity[21, 14, 12, 22, 5] Construct-validity critiques emphasize that benchmark scores may reflect a confounded mixture of capability, dataset-specific optimization, and artifacts such as contamination or item-level design flaws. We show that such confounded signals need not undermine predictive validity for downstream adoption: when adopters respond to the same observable signals embedded in benchmarks, these confounds can be shared across evaluation and selection processes, producing meaningful score–adoption correlations even in the absence of clean measurement of latent capability. This implies that predictive validity can arise from alignment between benchmark structure and adoption incentives rather than from faithful measurement of underlying ability. In this sense, construct validity and predictive validity are decoupled properties: a benchmark may fail as a pure capability measure while remaining informative about which models will be adopted. This suggests that benchmark signal is not an intrinsic property of the evaluation alone, but emerges from a time-dependent interaction between evaluation design and the surrounding ecosystem of model development, release incentives, and adoption (including market and visibility dynamics).

Limitations and Future work. Before drawing implications, we flag what this finding does not establish. We control for parameter count and a binary frontier-lab indicator; under joint control, four of five OLLM v2 benchmarks retain significant positive rank correlation with adoption in the 2024 cohort, while IFEval attenuates to borderline significance. We do not control for ecosystem maturity, training-data overlap with evaluation items, or finer-grained organizational reputation. Our analysis is restricted to open-weight models scored on a single leaderboard, with per-cohort sample sizes of $n = 11$-24 that limit statistical power; citation-based adoption captures research use, not deployment or end-user uptake. The analysis is also conditioned on OLLM's voluntary submission process: the matched analytic subset of 63 models includes only models submitted to the leaderboard, so the estimated score-adoption relationship may not generalize to all open-weight LLMs, particularly models from less represented labs, regions, or application domains. Finally, Spearman correlation establishes monotonic rank agreement rather than linear dose-response or causality. Future work should evaluate additional leaderboards and prospective cohorts.

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A Adoption Classification Details

A.1 Pipeline

For each citing paper $i \in S_m$, we extract citation sentences referencing model m with a three-sentence context window (the citation sentence, one sentence before, and one sentence after). Models are cited in 1.05 sentences per paper on average, so paper-level and sentence-level adoption labels coincide in practice, and we apply sentence-level classifier error rates at the paper level. We classify each sentence as *adoption* (use, fine-tuning, or extension) or *context* (background or comparative reference) via zero-shot prompting of GPT-4.1-mini [40]. For citations to publications introducing multiple variants (e.g., Llama 7B and 70B), we disambiguate the cited variant using Llama-3.1-8B [19].

A.2 Validation performance

We evaluate the classifier on a held-out set of 300 manually annotated citation sentences, with true adoption prevalence 24.9% (consistent with the population rate of 25.4% estimated from a separate annotated sample). Per-class performance is reported in Table 2.

Table 2: Classifier performance on validation set.

Class	Precision	Recall	F1
Adoption	0.760	0.576	0.656
Context	0.870	0.940	0.904
Macro-F1		0.780	
Accuracy		0.849	

The classifier under-predicts adoption: predicted adoption rate 18.9% versus true rate 24.9%, an absolute undercount of 6.0 percentage points. This bias motivates the ACC correction in Equation 1.

A.3 Prompts

We used the following prompt for classification with GPT-4.1-mini:

System Instruction

Prompt: *You are an expert in many areas of the scientific literature, with a specialty in how machine learning models are used.*

Prompt Template

Prompt: *You will pretend to be the author of some sentences from an academic paper which reference {model_descriptor}. Your goal is to determine the extent to which you used the cited model in your paper. The specific citation which references the foundation model is highlighted using HTML style* $\langle$*cite*$\rangle$ *brackets.*

Choose from the following list to determine the extent to which you adopted the model:

(1) I merely referenced the cited model as relevant background, for its methodology, or its dataset.
(2) I use the cited model or a part of the model itself, but I don't alter the weights.
(3) I make updates to the cited model's weights, through additional gradient based training such as fine-tuning.

Please respond in JSON format only, with key "answer" and value either 1, 2, or 3.

The sentences are as follows:

"{multisentence}"

B Adjusted Classify-and-Count

We adapt the Adjusted Classify-and-Count (ACC) methodology [23], which has recently been applied to estimating LLM-use prevalence in crowd work [53]. Naive counts of predicted adoptions are biased whenever classifier recall and specificity are imperfect, and the magnitude of the bias depends on each model's true adoption prevalence, distorting cross-model rankings. Let $\hat{p}_m^{\text{pred}}$ denote the predicted adoption rate among retrieved papers S_m. Under the assumption that classifier error rates are stable across models, defensible here because our population is restricted to LLMs released in 2023–2024, the unbiased estimator of the true adoption rate is

$$\hat{p}_m^{\text{true}} = \text{clip}_{[0,1]}\left(\frac{\hat{p}_m^{\text{pred}} - \widehat{\text{FPR}}}{\widehat{\text{TPR}} - \widehat{\text{FPR}}}\right). \tag{1}$$

C Inverse-Probability Weighting

The retrievable subset $S_m \subseteq C_m$ is non-random: papers indexed in S2ORC, posted to arXiv, or available through open-access channels are over-represented. To generalize to the full citing population C_m, we estimate retrievability propensities $\hat{\pi}_i = \Pr(i \in S_m \mid \mathbf{x}_i)$ by logistic regression on venue, publication year, open-access status, and field-of-study, fitted on the complete S2AG citation graph for the models in our study.

The Hájek estimator of the predicted adoption rate is

$$\hat{p}_m^{\text{pred,IPW}} = \frac{\sum_{i \in S_m} w_i \hat{y}_{im}}{\sum_{i \in S_m} w_i}, \quad w_i = 1/\hat{\pi}_i, \tag{2}$$

which we substitute into Equation 1 to obtain $\hat{p}_m^{\text{true,IPW}}$ and the population-scaled adoption count

$$\widehat{A}_m = \hat{p}_m^{\text{true,IPW}} \cdot N_m. \tag{3}$$

To guard against high-variance weights from low-propensity papers, we trim weights at the 99th percentile.

D Uncertainty Quantification

We propagate two sources of uncertainty into $\widehat{A}_m$:

1. **Sampling variability.** For each bootstrap iteration b, we draw n_m papers from S_m with replacement, retaining each sampled paper's predicted label $\hat{y}_{im}$ and retrievability weight w_i jointly. We recompute the Hájek predicted rate on the resampled set:

$$\hat{p}_m^{\text{pred,IPW},(b)} = \frac{\sum_{i \in S_m^{(b)}} w_i \hat{y}_{im}}{\sum_{i \in S_m^{(b)}} w_i}. \tag{4}$$

   Resampling papers (rather than drawing the count a_m from a Binomial) is necessary under IPW: the variance of $\hat{p}_m^{\text{pred,IPW}}$ depends on the empirical weight distribution through the effective sample size $n_m^{\text{eff}} = (\sum_i w_i)^2 / \sum_i w_i^2 \leq n_m$, which a parametric Binomial draw on the count cannot represent.

2. **Classifier calibration.** TPR and FPR are estimated from a finite validation sample. We model their posteriors as independent Beta distributions derived from the validation confusion matrix:

$$\widehat{\text{TPR}} \sim \text{Beta}(\text{TP}+1,\, \text{FN}+1), \qquad \widehat{\text{FPR}} \sim \text{Beta}(\text{FP}+1,\, \text{TN}+1). \tag{5}$$

For $b = 1, \ldots, B$ with $B = 10{,}000$, we draw $(\widehat{\text{TPR}}^{(b)}, \widehat{\text{FPR}}^{(b)})$ once per iteration (shared across all models, reflecting the shared classifier) and resample papers within each model independently. We compute $\hat{p}_m^{\text{true,IPW},(b)}$ via Equation 1 followed by $\widehat{A}_m^{(b)} = \hat{p}_m^{\text{true,IPW},(b)} \cdot N_m$. Point estimates of $\widehat{A}_m$ are computed once on the original S_m with observed weights; the bootstrap propagates spread only. We report the median of the bootstrap distribution (which agrees with the point estimate to within Monte Carlo error) as $\widehat{A}_m$ and the 2.5^{th} and 97.5^{th} percentiles as 95% intervals.

The variance of $\hat{p}_m^{\text{true}}$ is inflated relative to $\hat{p}_m^{\text{pred}}$ by a factor of $(\widehat{\text{TPR}} - \widehat{\text{FPR}})^{-2} \approx 3.75$, which is the unavoidable cost of unbiased quantification under imperfect classification. This inflation is reflected in the bootstrap intervals.

For downstream rank correlations between $\widehat{A}_m$ and benchmark scores, we use Spearman ρ on the bootstrap-median $\widehat{A}_m$, with permutation tests ($N_{\text{perm}} = 10{,}000$, two-sided) for significance and row-resampling bootstraps ($N_{\text{boot}} = 10{,}000$) for 95% confidence intervals on ρ. Permutation p-values use the Phipson-Smyth $(k+1)/(N_{\text{perm}}+1)$ correction [41]. Where we control for model size, we report partial Spearman correlations with residual permutation p-values [34, 4]. Specifically, we regress $R(y_i)$ and $R(S_{b,i})$ on $R(z_i)$, permute the resulting rank-residuals, and recompute $\rho_{s,a|z}^b$ for $N_{\text{perm}} = 10{,}000$ draws. This procedure is asymptotically equivalent to the Freedman–Lane

scheme for testing partial correlations under linear residual structure [4]. For double control (size and frontier-lab indicator), $R(S_{b,i})$ and $R(y_i)$ are residualized jointly on $(R(z_i), f_i)$ and the Pearson correlation of residuals is computed and permuted analogously.

E Sensitivity analysis

We report the trimming robustness check. Headline results are stable in sign and significance.

Benchmark	**Cohort**	**N**	$\rho_{s,a}$	**95% CI**	q_ρ
IFEval (2023)	2024	23	0.58**	[0.22, 0.80]	0.01
BBH (2022)	2023	17	0.18	[-0.36, 0.65]	0.56
	2024	23	0.56**	[0.19, 0.79]	0.01
MATH Lvl 5 (2021)	2023	17	-0.12	[-0.58, 0.37]	0.65
	2024	22	0.61**	[0.21, 0.87]	0.01
GPQA (2023)	2024	23	0.60**	[0.29, 0.78]	0.01
MuSR (2023)	2024	23	0.71***	[0.42, 0.87]	0.00

Table 3: Benchmark–adoption correlations after trimming the highest- and lowest-scoring model. Outcome: log cumulative adoptions, one year post-release. $\dagger p < 0.10$, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.